DAY-1: Foundations

1.LLM:

LLM stands for large language model. It is an AI System that trained on large amount of textual Data. It learns patterns from text and predicts most likely next word to generate human like responses. It does not understand the things like human. it only predicts the next word based on their training.

a. **Why is It Useful? →** LLMs can understand instructions written in natural language and generate useful responses. This can reduce the effort required for tasks such as writing, coding, summarizing, translating, and answering questions.
b. **Real-World Use Case →** A software developer could use ChatGPT to write code, summarize error messages, and create documentation.
c. **Experiment→**

   Prompt: Explain Artificial Intelligence in 50 words.

Figure 1.

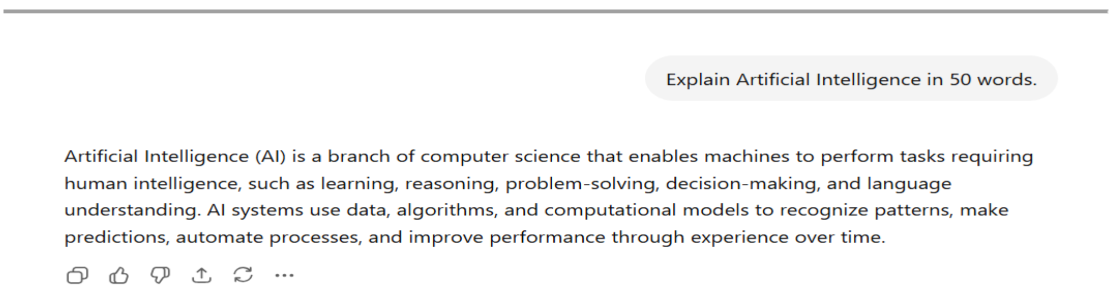
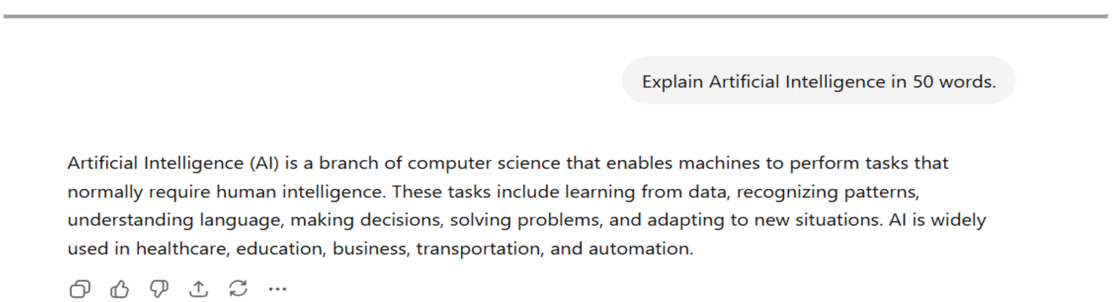


Figure 2.



d. **Observation→** If the question is repeated, the word of the answer will vary but the meaning is the same. This is because the model can select other subsequent words to predict and sentence structure.

e. **Conclusion→** LLMs "infer" answers to a question by guessing what the next word or phrase in the context might be and finding a pattern they have learned. The different possible predictions yield different answers.

2.Prompts:

A prompt is set of instruction or input given to an LLMs. The quality of the prompt directly affects the quality of the output.

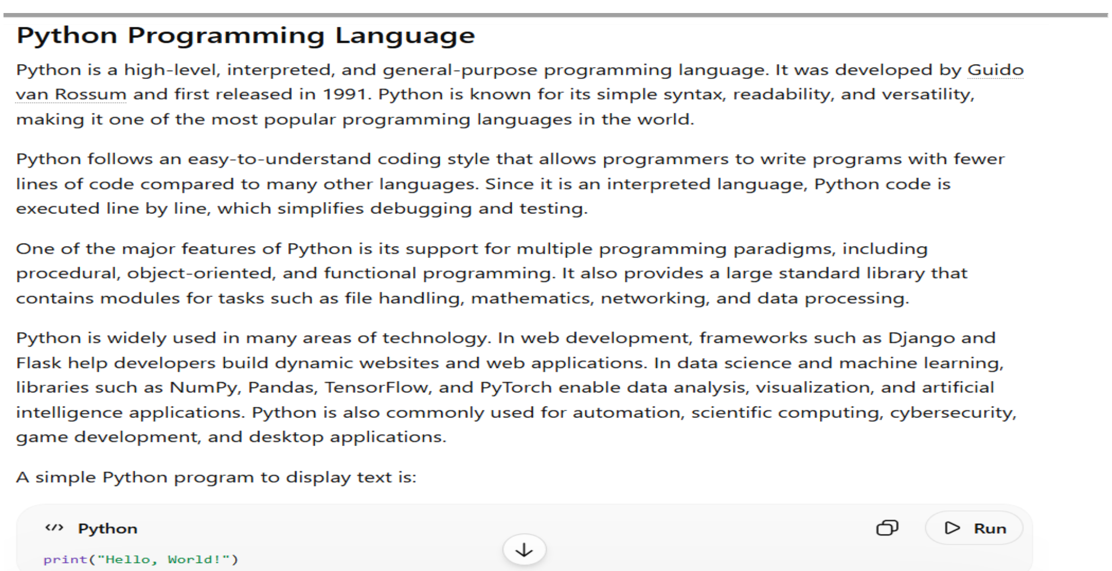
a. **Why is It Useful? →** A well-written prompt provides clear guidance, helping the model generate accurate and relevant responses.

b. **Real-World Use Case →** Companies use carefully designed prompts in AI chatbots to provide customer support.

c. **Experiment→**

Weak Prompt: Explain Python in one page.

Figure 1.



Python Programming Language

Python is a high-level, interpreted, and general-purpose programming language. It was developed by Guido van Rossum and first released in 1991. Python is known for its simple syntax, readability, and versatility, making it one of the most popular programming languages in the world.

Python follows an easy-to-understand coding style that allows programmers to write programs with fewer lines of code compared to many other languages. Since it is an interpreted language, Python code is executed line by line, which simplifies debugging and testing.

One of the major features of Python is its support for multiple programming paradigms, including procedural, object-oriented, and functional programming. It also provides a large standard library that contains modules for tasks such as file handling, mathematics, networking, and data processing.

Python is widely used in many areas of technology. In web development, frameworks such as Django and Flask help developers build dynamic websites and web applications. In data science and machine learning, libraries such as NumPy, Pandas, TensorFlow, and PyTorch enable data analysis, visualization, and artificial intelligence applications. Python is also commonly used for automation, scientific computing, cybersecurity, game development, and desktop applications.

A simple Python program to display text is:

Python

Run

```python
print("Hello, World!")
```

Good Prompt: Act as a Python teacher. Write a one-page explanation of Python for beginners. Include Introduction, Features, Advantages, Applications, Syntax Examples (variables, loops, functions), and Conclusion. Use clear headings, simple language, and practical examples.

Figure 2.

Python Programming Language

Introduction

Python is a high-level, interpreted, and general-purpose programming language created by Guido van Rossum in 1991. It is known for its simple syntax and readability, making it ideal for beginners.

Features

- Easy to learn and use
- Interpreted language
- Object-oriented
- Cross-platform support
- Large standard library

Advantages

- Simple and readable code
- Faster development
- Free and open source
- Strong community support

Applications

- Web Development
- Data Science
- Artificial Intelligence
- Automation
- Game Development

Syntax Examples

Variable

Python Run

```python
name = "Harsh"
print(name)
```

Loop

Python Run

```python
for i in range(5):
    print(i)
```

Loop

Python

```python
for i in range(5):
    print(i)
```

Function

Python

```python
def greet():
    print("Hello")

greet()
```

Conclusion

Python is one of the most popular programming languages because it is easy to learn, powerful, and widely used in various fields such as web development, AI, data science, and automation. It is an excellent choice for beginners and professionals alike.

d. **Observation→** The improved prompt provides:

- Better detail
- Better structure
- More relevant information

e. **Conclusion→** Specific prompts produce better results than simple and vague prompts.

3. Tokens:

A token is the basic unit processed by an LLM. A token may be a word, part of a word, punctuation mark, or symbol.

Example: I love learning AI.

Possible token split:

- **Token 1.** I
- **Token 2.** love
- **Token 3.** learning
- **Token 4.** AI

➢ **Token 5. “.”**

In above example words are 4 but tokens are 5.

a. **Why is It Useful? →** LLMs process text as tokens, not words. Token count determines:
   - ➢ Processing cost
   - ➢ Input limits
   - ➢ Output limits

b. **Real-World Use Case →** API providers take charges based on the number of tokens processed by user.

c. **Observation→** Words and tokens are not always equal. Long words may be split into multiple tokens

d. **Conclusion→** Understanding tokens helps manage LLM costs and context limits effectively.

4. Context Window:

The context window is the amount of information an LLM can remember and use while generating a response in a conversation and LLM use that information as context to generate Future responses.

a. **Why is It Useful? →** It allows the model to remember previous messages and maintain conversation continuity.

b. **Real-World Use Case →** Customer support chatbots use context windows to remember earlier customer issues during a conversation

c. **Experiment→**

   Prompt 1: Remember this number: 987654

Figure 1.

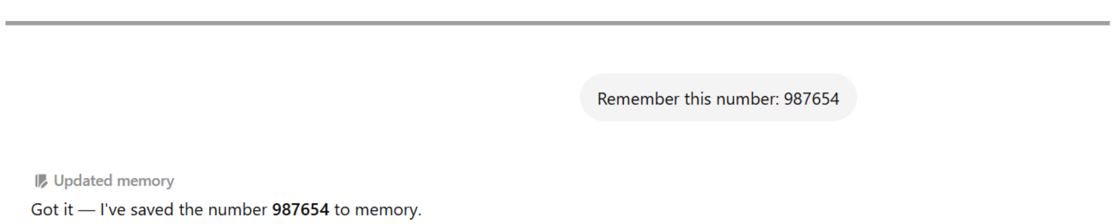
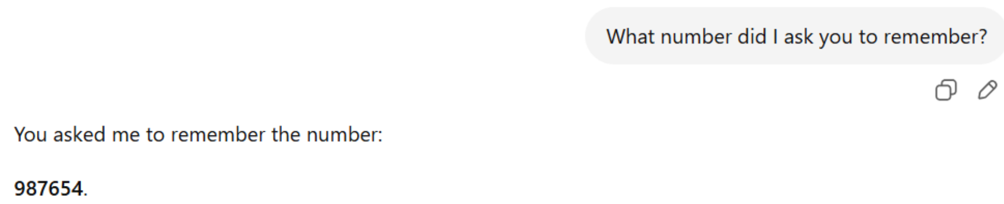


Figure 2.



d. **Observation→** If the conversation becomes too long and exceeds the context window, the model may forget earlier information.

e. **Conclusion→** The context window acts like the model's working memory. Larger context windows allow the model to remember more information.

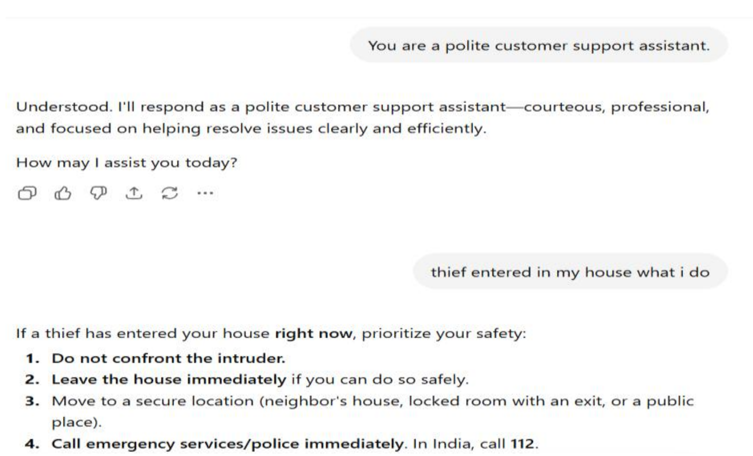
DAY-2: Controlling the Model

1. System Prompt vs User Prompt:

A **System Prompt** sets the model's overall behaviour, role, and rules. A **User Prompt** is the instruction given by the user during the conversation.

a. **Why is it useful? →** System prompts help control how the AI responds, while user prompts specify what task needs to be performed.

b. **Real-World Use Case →** A customer support chatbot may use a system prompt: **Prompt**: You are a polite customer support assistant.



Experiment→
User Prompt: Explain Machine Learning.

Explain Machine Learning in 100 words

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed. Instead of following fixed rules, ML algorithms identify patterns, make predictions, and adapt based on experience. It is widely used in applications such as email spam filtering, recommendation systems, image recognition, speech processing, fraud detection, and self-driving cars. Machine learning typically involves collecting data, training a model, testing its accuracy, and using it to make decisions on new data. ML helps automate tasks, improve efficiency, and solve complex real-world problems.

System Prompt: You are a college professor. Explain concepts in simple language.

You are a college professor. Explain concepts in simple language and explain machine learning in 100 words

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make decisions without being explicitly programmed for every situation. It works by analyzing examples, identifying patterns, and using those patterns to predict outcomes or solve problems. For example, a machine learning model can learn to recognize spam emails by studying thousands of labeled emails. ML is widely used in recommendation systems, image recognition, speech assistants, fraud detection, and healthcare. The more relevant data a model receives, the better it can learn and improve its accuracy, making technology smarter and more efficient.

Observation→Without a system prompt, the answer is general.
With a professor role, the answer becomes more educational and detailed.

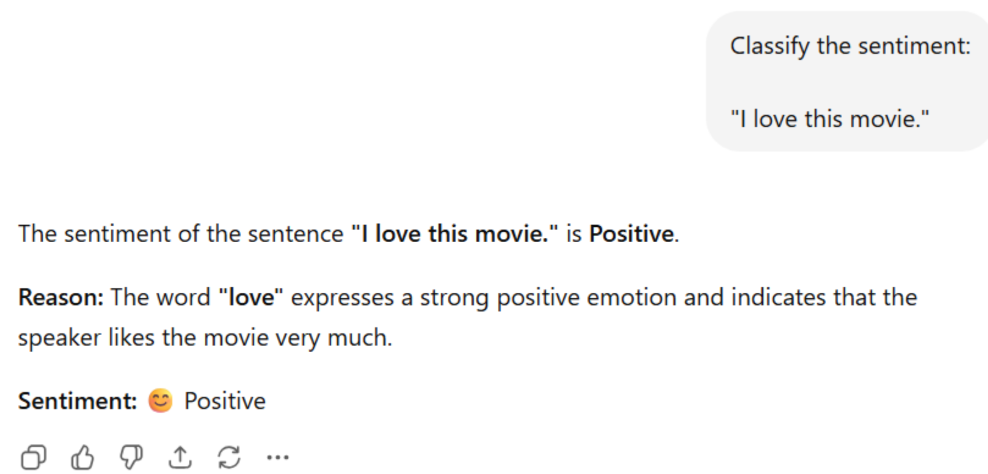
Conclusion→System prompts influence the AI's personality, behaviour, and response style, while user prompts define the task.

2. Zero-Shot Prompting:

Zero-shot prompting means asking the model to perform a task without providing any examples.

a. **Why is it useful? →** It is simple, fast, and works well for common tasks the model already understands.
b. **Real-World Use Case ➔**Classifying customer feedback as positive or negative.

Experiment:
Prompt: Classify the sentiment: "I love this movie."



c. **Observation➔**The model correctly identifies the sentiment even without examples.
d. **Conclusion➔** Zero-shot prompting is useful when the task is simple and already familiar to the model.

3. Few-Shot Prompting:

Few-shot prompting provides a few examples before asking the model to perform a task.

a. **Why is it useful? →** Examples help the model understand the expected format and behaviour.
b. **Real-World Use Case ➔** Extracting information from invoices in a consistent format.

Experiment:

Text: "I love this movie"
Sentiment: Positive

Text: "This is terrible"
Sentiment: Negative

Text: "Amazing experience"
Sentiment:

Text: "I love this movie"
Sentiment: Positive

Text: "This is terrible"
Sentiment: Negative

Text: "Amazing experience"
Sentiment: Positive

c. **Observation➔** The model follows the pattern shown in the examples and produces more consistent results.

d. **Conclusion➔** Few-shot prompting works better than zero-shot when tasks require specific formats or complex reasoning.

4. Chain of Thought (Cot):

Chain of Thought prompting encourages the model to solve a problem step by step rather than jumping directly to the answer.

a. **Why is it useful? →** Breaking a problem into smaller steps often improves reasoning accuracy.
b. **Real-World Use Case ➔** Mathematical calculations, logical reasoning, and multi-step decision making.

Experiment:
Prompt 1:

A train travels at 60 km/h for 3 hours.
How far does it travel?

180 km

Prompt 2:

A train travels at 60 km/h for 3 hours.
Think step by step and calculate the distance.

Let's calculate it step by step:

1. **Speed of the train** = 60 km/h
2. **Time traveled** = 3 hours
3. **Distance = Speed × Time**

$$\text{Distance} = \text{Speed} \times \text{Time}$$

So,

Distance = 60 × 3 = 180 km

✅ **Answer: The train travels 180 kilometers.** 🚆

c. **Observation➔** The step-by-step approach makes the reasoning process clear and often reduces mistakes

d. **Conclusion➔** Chain of Thought improves performance on reasoning-heavy tasks by encouraging logical intermediate steps.

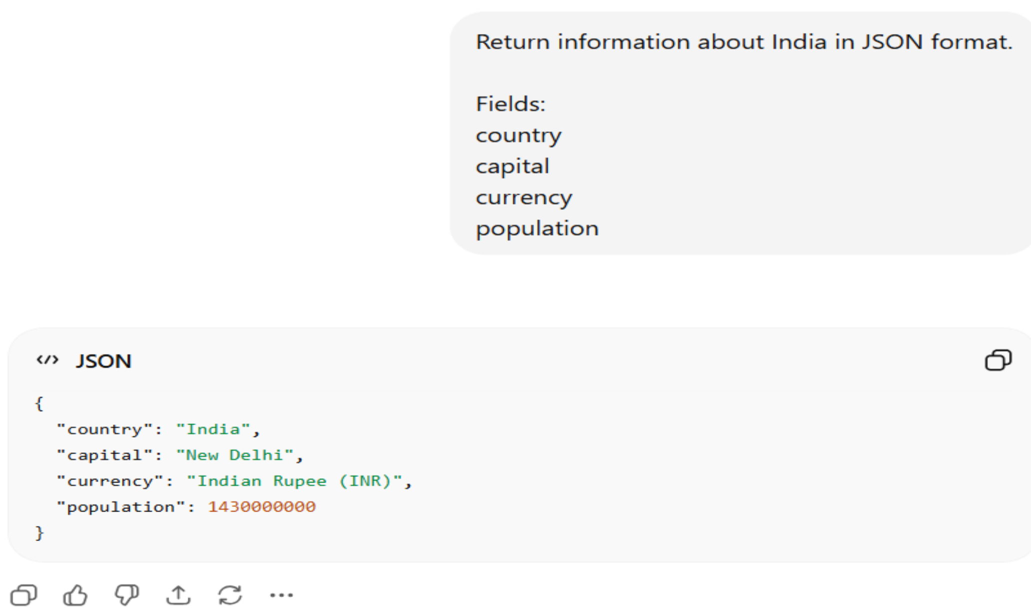
5. Structured Outputs / JSON Responses:

Structured output means asking the model to return information in a predefined format such as JSON.

a. **Why is it useful? →** JSON can be easily processed by software applications, APIs, databases, and automation systems.

b. **Real-World Use Case ➔** AI extracting customer information and sending it directly to a CRM system.

Experiment:



c. **Observation➔** The output can be directly consumed by programs without additional parsing.

d. **Conclusion➔** Structured outputs make LLMs practical for real applications where data must be processed automatically.

DAY 3: Beyond a Single Model

1. Tool Calling:

Tool calling allows an LLM to use external tools such as calculators, databases, APIs, web search, or code interpreters to perform tasks it cannot reliably do on its own.

a. Why is it useful? → LLMs are good at language but not always reliable for:

- Complex calculations
- Real-time information
- Database lookups
- External actions
- Tools provide accurate and up-to-date information.

b. Real-World Use Case → A travel chatbot checking live flight prices through an API.

Experiment→

User Prompt:

> What is 56789 × 98765?

Step 1: Detect the task
The model recognizes that this is a mathematical calculation.

Step 2: Call the calculator tool
Instead of multiplying internally, it sends the expression to the calculator:

```
calculator("56789 * 98765")
```

Step 3: Tool executes
The calculator computes:

```
5608765585
```

Step 4: Tool returns result
The calculator sends the answer back to the model.

Step 5: Model responds

```
56789 × 98765 = 5,608,765,585
```

c. Observation→ The calculator gives an exact result every time, while an LLM may occasionally make arithmetic mistakes.

d. Conclusion→ Tool calling combines the language ability of an LLM with the accuracy of specialized tools.

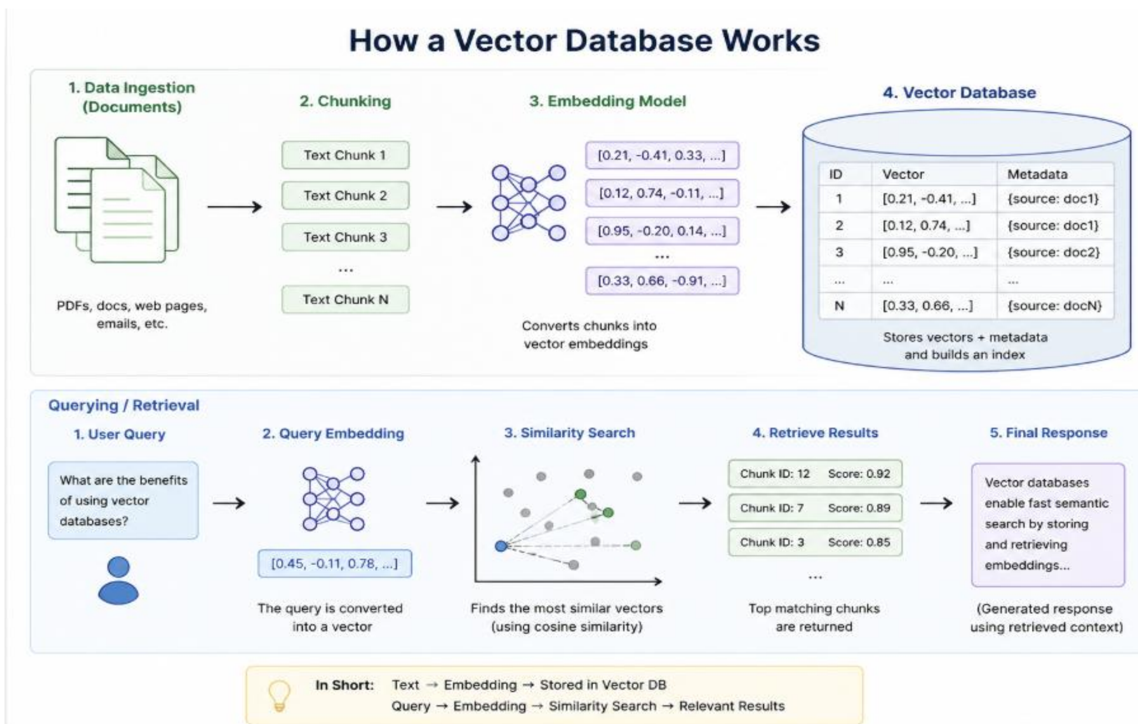
2. Vector Database:

A vector database stores information as numerical embeddings (vectors) that represent meaning. Instead of searching for exact words, it searches for similar meanings.

a. Why is it useful? → LLMs are good at language but not always reliable for:

- Enables semantic search.
- Retrieves relevant information using meaning rather than keywords.
- Supports Retrieval-Augmented Generation (RAG).
- Handles large document collections efficiently.

b. Working of Vector Database:→



c. **Observation→** Vector search finds similar content because the meaning is similar.

d. **Conclusion→** Vector databases help LLMs find relevant information based on meaning rather than exact words.

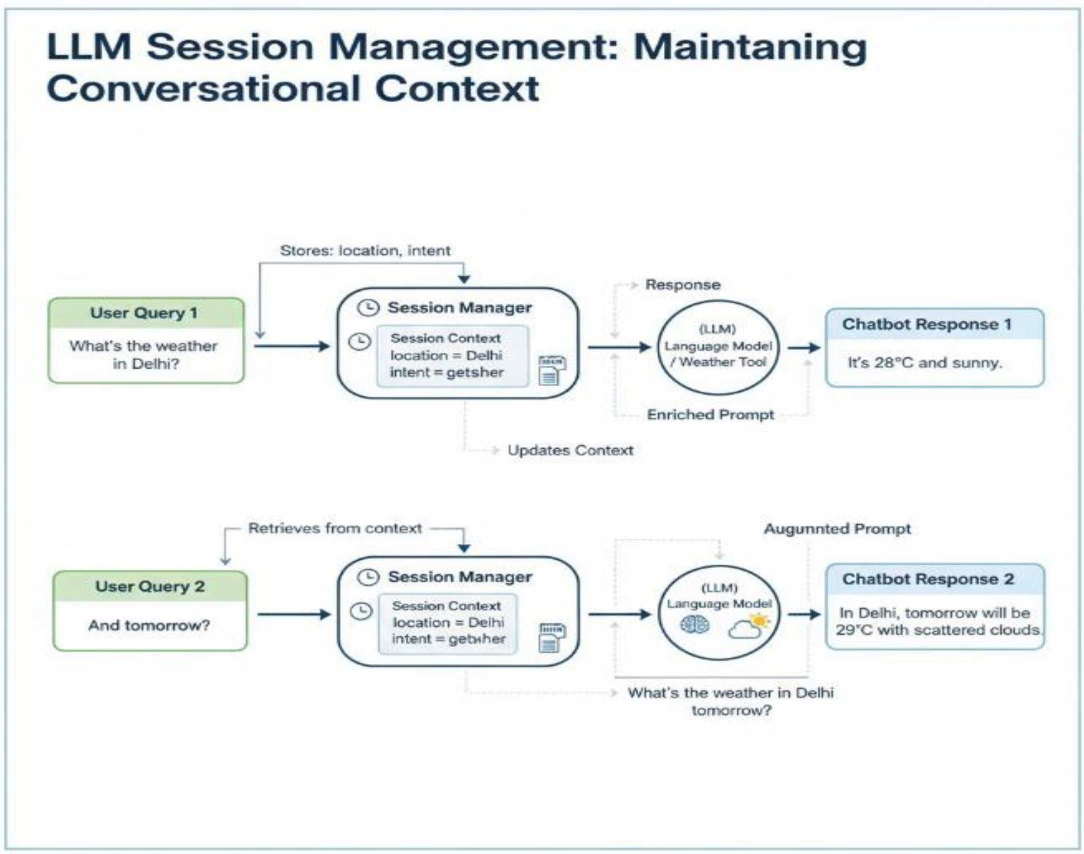
3. Session / Memory Management:

- ➢ Session management allows an AI system to remember previous messages and maintain conversation context.
- ➢ Memory management determines what information should be retained, summarized, or forgotten.

a. **Why is it useful? →** Without memory, every message would be treated as a completely new conversation.

b. **Real-World Use Case →** A customer support chatbot remembering a user's issue across multiple messages.

Working→



c. **Observation→** The model can maintain context and refer to earlier messages when session information is available.

d. **Conclusion→** Tool calling combines the language ability of an LLM with the accuracy of specialized tools

4. Agent vs LLM:

An **LLM (Large Language Model)** is an AI model designed to understand and generate human-like text. It takes an input, processes it, and produces a response. An **Agent** is a system built on top of an LLM that can use tools, maintain memory, plan multiple steps, and take actions to achieve a goal. While an LLM mainly generates answers, an Agent can reason, act, and interact with external systems.

a. Why is it useful? →

LLMs are powerful for language tasks but have limitations:

- They typically perform one task at a time.
- They cannot reliably access real-time information on their own.
- They do not naturally plan multi-step workflows.
- They cannot directly interact with external systems.

Agents overcome these limitations by:

- Using tools such as APIs, databases, and web search.
- Maintaining memory across tasks.
- Planning and executing multiple steps.
- Working toward a specific goal.

b. Real-World Use Case →

A customer asks: *"Book the cheapest flight from Delhi to Mumbai for next Friday and send me the details."*

- An LLM can explain how to search for flights, but an Agent can:
- Search flight websites.
- Compare prices.
- Select the cheapest option.
- Store the results.
- Provide the final recommendation.

Experiment:

LLM Task:
Prompt: Create a 3-day Goa travel itinerary.
Output: The LLM generates a travel plan based on its knowledge.
Agent Task

Agent Task:

Prompt: Find flights to Goa, compare hotel prices, create a travel itinerary, and recommend the cheapest option.

Output: The Agent can use tools, gather information, compare options, and complete multiple steps before producing the final answer.

c. **Observation→** The LLM is effective at generating text and answering questions, but it is limited to the information available in its context. An Agent can use tools, remember information, and perform multiple actions, making it suitable for solving more complex real-world problems.

d. **Conclusion→** An LLM is the intelligence that understands and generates language, whereas an Agent combines that intelligence with memory, planning, and tools to accomplish goals. In simple terms, an **LLM can answer questions, while an Agent can take actions and solve multi-step tasks.**